

# **JNTU Hyderabad**

## **MODEL QUESTION PAPER (R20 PATTERN)**

Model Question Paper 1 ? Engineering Mechanics (Latest Pattern)

Subject Code: ME105 Semester: 2 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Statics of Particles)**

1. Write short notes on Statics of Particles. [12 Marks]
2. List the advantages and limitations of Statics of Particles. Explain any two in detail. [12 Marks]
3. List the advantages and limitations of Statics of Particles. Explain any two in detail. [12 Marks]

### **SECTION B (Unit 2: Equilibrium of Rigid Bodies)**

4. Write short notes on Equilibrium of Rigid Bodies. [12 Marks]
5. Discuss the engineering applications of Equilibrium of Rigid Bodies. [8 Marks]
6. Derive the expression related to Equilibrium of Rigid Bodies and explain each term. [8 Marks]

### **SECTION C (Unit 3: Centroid and Moment of Inertia)**

7. Compare and contrast the key concepts involved in Centroid and Moment of Inertia. [12 Marks]
8. Derive the expression related to Centroid and Moment of Inertia and explain each term. [8 Marks]
9. How is Centroid and Moment of Inertia implemented? Give an algorithm / procedure and explain. [8 Marks]

### **SECTION D (Unit 4: Dynamics of Particles)**

10. Define Dynamics of Particles. Explain its significance in Engineering Mechanics with a suitable example. [12 Marks]
11. Solve a problem related to Dynamics of Particles. Show all the steps clearly. [12 Marks]
12. Derive the expression related to Dynamics of Particles and explain each term. [12 Marks]

### **SECTION E (Unit 5: Friction and Virtual Work)**

13. Derive the expression related to Friction and Virtual Work and explain each term. [12 Marks]
14. How is Friction and Virtual Work implemented? Give an algorithm / procedure and

**explain. [12 Marks]**

15. Describe the working principle of Friction and Virtual Work with necessary equations. [12 Marks]

\*\*\* End of Question Paper \*\*\*